Luxin Gabriel Mătăsaru

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Education

• École Polytechnique *Sept 2024 - June 2027*

Bachelor's degree in Mathematics and Computer Science GPA 4.19/4.3

• Tudor Vianu National High School of Computer Science *Sept 2020 - June 2024*

Baccalaureate Diploma 9.73 (Romanian, Mathematics, Physics)

Experience

• Arago, internship - Software Team *June - August 2026*

- Implementing a compute dispatcher on a cluster of ARM and RISCV embedded cores, fully written in baremetal C and C++.
- Debugged low level device drivers.
- Implemented compilation, simulation and testing infrastructure for my project.

• Arago, internship - Optics Team *June - August 2025*

- Modeling time response of the up-conversion phenomenon. Correlated simulation to experimental results. Showed how the system parameters could be tuned to go from response time of **s** to an order of μ**s**.
- Improved prototyping efficiency by creating a Python module for automatic data acquisition through UART and I^2C, using the optics lab equipment.

• Black Shell, internship *June-July 2023*

- One of the first full time employees at the beginning of the startup.
- Learned Golang and used it to implement key manager APIs for Wireguard and OpenVPN which shiped with the company's main product.

Extracurriculars

• IP Paris Formula Student Racing - Electronics Team *2024-present*

- Working on the Break System Plausibility Device (BSPD), a non-programmable, system critical device that shuts down power when breaking and accelerating at the same time.
- Designed the "ready to drive" circuit, responsible for powering on the car.
- Ensured both boards fit within the rule's specifications.

• CSC - CS Committee of the Polytechnique Bachelor *2025-2026*

- Vice President; coordinating a 10-member team to organize seminars, competitions, hackathons and other events.
- Presented an "Introduction to Diffusion" (*link*), a seminar aimed at giving some intuition for how diffusion works while also going into the mathematics of DDPM.

Projects

• High Redshift Galaxy Clustering *2026*

Using unsupervised machine learning to find hidden structure within high redshift galaxies

- Reproduced analysis performed by **Tohill et al.**.
- Extended the paper by including the **COSMOS** dataset.
- Was responsible for analyzing the latent space, using dimensionality reduction (UMAP, t-SNE) and clustering (K-Means, Hierarchical Clustering).
- Identified clusters correlated well with physical parameters.

• Implementation of QR-DQN for CleanRL *2026*

Lab Reserch Project under the supervision of Dr. Simo Alami

- Implemented the Quantile Regression DQN algorithm (**Dabney et al. 2017**), and contributed it to the CleanRL open source library.
- Matched the performance reported in the original paper on 5 games of the Atari 2600 benchmark.
- Wrote a report and held an oral defense explaining my work.

• Swarm Rescue Competition *2025*

Writing a controller to coordinate a swarm of drones in a simulated environment.

– Part of a team of 3 students participating in the **Swarm Rescue Competition**
– Designed a state machine controller for high level decision making
– Implemented PID controller for the drone's motion.
– Merged the simulated IMU and GPS measurements to reduce noise in drone position.
– **source code**

• **DiffEdit** *2025*

Implementation of an image editing method using diffusion.

– Made extensive use of the PyTorch and Hugging Face diffusers libraries to write my own diffusion loop as described in the **paper**.
– **source code**

• **NSS Space Settlement Contest** ***Grand Prize***

*Designed a self-sufficient space colony detailed in a **50 page article**.* *2024*

– Won the Grand Prize out of 1000+ teams.
– Presented at the 2024 International Space Development Conference.
– Responsible for the structural design and analysis.
– Created 3D models and renders in Blender and Fusion360.

• **Optics Unravelled** *2023*

Built an open source interactive optics simulator & lesson platform.

– Designed the website to be used as a teaching tool by my high school. (**opticsunravelled.com**)
– Full stack application in Node.js. (**source code**)

• **Polytechnic University of Bucharest Cube2Space National Contest** ***1st Prize***

Design of a pocket cube to be launched up to 9km on the LUNA rocket. *2022*

– As leader, managed and organised the ViaSpace team.
– The pocket cube investigated dampening methods to mitigate the effects of vibrations during launch.
– Created CAD models in Fusion360, used for rendering and 3D printing of the pocket cube.

• **ESA Moon Camp Pioneers** ***3rd Prize***

Design and 3D modeling of a lunar base in Fusion 360. *2021*

– In charge of designing the airlock, leisure and other areas of the moon base.
– Created 3D models and renders of these areas in Fusion360.

Honors & Awards

• **International Zautykov Olympiad (Physics)** - Bronze medal *2023*
• **National Physics Olympiad** - Silver Medal *2022*
• **National Informatics Olympiad** - Bronze Medal *2021 & 2022*
• **International Astronomy and Astrophysics Competition** - Bronze Medal *2022*
• **Unbreakable Hacking Challenge Romania (Team Section)** - 1st Prize *2022*
• **National Astronomy and Astrophysics Olympiad** - Honorable Mention *2021*

Skills

Languages: Romanian - native
English - C2
French - C1
German - A1
Computer: C / C++/ C# / Python/ Golang / JavaScript / HTML / Fusion360 / Blender / KiCad